



Lab Report - 04

Experiment on 04

Course Name: Structured Programming Lab

Course Code: CSE-122

Submitted to

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Lab Report-4

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Signature:

Date:

Problem-1: Write a C program to read the value of an integer m and display the value of n is 1 when m is larger than 0, 0 when m is 0 and -1 when m is less than 0. (Use switch-case)

Test Data :-5

Expected Output:

The value of n =-1

Source code:

```
re X Untitled 51.c X
1  #include<stdio.h>
2  int main()
3  {
4      int m;
5      scanf("%d",&m);
6      switch(m>0)
7      {
8          case 1:
9              printf("\nThe value of n = 1\n");break;
10         case 0:
11             switch(m<0)
12             {
13                 case 1:
14                     printf("\nThe value of n = -1\n"); break;
15                 case 0:
16                     printf("\nThe value of n = 0\n");break; }
17             }
18
19     return 0;
20
21 }
```

Output:

```
-5

The value of n = -1

Process returned 0 (0x0)   execution time : 4.272 s
Press any key to continue.
|
```

Problem-2: : Write a C program to find the minimum between two numbers using a switch-case.

Test Data :

10 20

Expected Output:

10

Source code:

```
Untitled 51.c
1  #include<stdio.h>
2  int main()
3  {
4      int num1,num2;
5      scanf("%d%d",&num1,&num2);
6      switch(num1<num2)
7      {
8          case 1:
9              printf("\n%d\n",num1);break;
10         case 0:
11             printf("\n%d\n",num2); break;
12     }
13     return 0;
14
15 }
```

Output:

```
10 20

10

Process returned 0 (0x0)   execution time : 3.414 s
Press any key to continue.
```

Problem-3:Write a C program to check even or odd numbers using a switch-case.

Test Data :

7

Expected Output:

7 is an odd number.

Source code:

```
here X  Untitled 51.c X
1  #include<stdio.h>
2  int main()
3  {
4      int num;
5      scanf("%d",&num);
6      switch(num%2==0)
7      {
8          case 1:
9              printf("\n%d is an even number.\n",num);break;
10         case 0:
11             printf("\n%d is an odd number.\n",num);break;
12     }
13
14     return 0;
15
16 }
```

Output:

```
7

7 is an odd number.

Process returned 0 (0x0)   execution time : 1.589 s
Press any key to continue.
|
```

Problem-4:Write a C program to check positive negative or zero using a switch-case.

Test Data :

-5

Expected Output:

-5 is a negative number.

Source code:

```
e X Untitled52.c X
1  #include<stdio.h>
2  int main()
3  {
4      int num;
5      scanf("%d",&num);
6      switch(num>0)
7      {
8          case 1:
9              printf("\n%d is a positive number.\n",num);break;
10         case 0:
11             switch(num<0)
12             {
13                 case 1:
14                     printf("\n%d is a negative number.\n",num);break;
15                 case 0:
16                     printf("\n%d is zero.\n",num);break;
17             }
18         }
19     return 0;
20 }
```

Output:

```
-5
-5 is a negative number.
Process returned 0 (0x0)   execution time : 10.071 s
Press any key to continue.
|
```

Problem-5: Write a program in C to read any Month Number in an integer and display the number of days for this month using a switch-case.

Test Data :

7

Expected Output:

There are 31 days in month 7.

Source code:

```
here X  Untitled 51.c X
1  #include<stdio.h>
2  int main()
3  {
4      int month;
5      scanf("%d",&month);
6      switch(month){
7          case 1: case 3: case 5: case 7: case 8: case 10: case 12:
8              printf ("There are 31 days in month %d.\n",month);break;
9          case 2:
10             printf("There is 28 or 29 days in month.\n");break;
11             case 4: case 6:case 9: case 11:
12                 printf("There are 30 days in month %d\n.",month);break;
13
14         default:
15             printf ("Invalid month.\n");
16         }
17         return 0;
18     }
```

Output:

```
7
There are 31 days in month 7.
.
Process returned 0 (0x0)   execution time : 1.408 s
Press any key to continue.
|
```

Problem-6: Write a C program to check whether an alphabet is a vowel or a consonant using a switch-case.

Test Data :

P

Expected Output:

P is a consonant.

Source code:

```
re X Untitled 51.c X
1  #include<stdio.h>
2  int main()
3  {
4      char ch;
5      scanf("%c",&ch);
6      switch(ch)
7      {
8          case 'a': case 'e': case 'i': case 'o': case 'u':
9              printf("\n%c is a vowel.\n",ch);break;
10         default:
11             printf(" \n%c is a consonant.\n",ch);
12         }
13         return 0;
14     }
```

Output:

```
p
p is a consonant.
Process returned 0 (0x0)   execution time : 2.008 s
Press any key to continue.
|
```


Problem-7: Write a C program to calculate profit and loss on a transaction using a switch-case.

Test Data :

Enter buying price: 500

Enter selling price: 300

Expected Output:

Loss amount: 200

Source code:

```
re x  Untitled52.c x
1  #include<stdio.h>
2  int main()
3  {
4      int buy,sell;
5      printf("Enter buying price:");
6      scanf("%d",&buy);
7      printf("Enter selling price:");
8      scanf("%d",&sell);
9      switch (buy>sell)
10     {
11     case 1:
12         printf("\nLoss amount:%d\n",buy-sell);break;
13     case 0:
14         sell-buy;
15         printf("profit amount:%d\n",sell-buy);break;
16     }
17     return 0;
18 }
```

Output:

```
Enter buying price:500
Enter selling price:300

Loss amount:200

Process returned 0 (0x0)   execution time : 5.670 s
Press any key to continue.
|
```

Problem-8: Write a program in C to convert temperature from Fahrenheit to Celsius and Celsius to Fahrenheit using a switch-case.

Test Data :

Press c to convert the temperature from Fahrenheit to Celsius.

Press f to convert temperature from Celsius to Fahrenheit.

Enter your choice (c, f): f

Enter temperature in Celsius: 37

Expected Output :

Temperature in Fahrenheit: 98.60

Source code:

```
re  x Untitled52.c  x  Untitled2.c  x
1  #include <stdio.h>
2  int main() {
3      char choice;
4      float c, f;
5      printf("Press c to convert temperature from Fahrenheit to Celsius.\n");
6      printf("Press f to convert temperature from Celsius to Fahrenheit.\n");
7      printf("Enter your choice (c, f): ");
8      scanf(" %c", &choice); // Skip newline with space before %c
9      switch (choice) {
10         case 'c':
11             printf("Enter temperature in Fahrenheit: "); scanf("%f", &f);
12             c = (f - 32) * 5 / 9;
13             printf("Temperature in Celsius: %.2f\n", c); break;
14         case 'f':
15             printf("Enter temperature in Celsius: "); scanf("%f", &c);
16             f = (c * 9 / 5) + 32;
17             printf("Temperature in Fahrenheit: %.2f\n", f); break;
18         default:
19             printf("Invalid choice. Please enter only 'c' or 'f'.\n");
20     }
21     return 0;
22 }
```

Output:

```
Press c to convert the temperature from Fahrenheit to Celsius.
Press f to convert temperature from Celsius to Fahrenheit.
Enter your choice (c, f): f
Enter temperature in Celsius: 37

Temperature in Fahrenheit: 98.60

Process returned 0 (0x0)   execution time : 4.452 s
Press any key to continue.
```